

Quiz 1 [25min]

Correct Answer 4pts, No Answer -1 pts, Wrong Answer -2 pts. If the total points < 0, your score will be set to 0.

**Check *ONLY* T if the given statement in the question is correct or only F if not.
If you find XXX, write the correct replacement for XXX.**

1. The universal quantification of $P(x)$ is the statement “ $P(x)$ for all values of x .” The notation $\forall xP(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the universal quantifier. We read $\forall xP(x)$ as “for all $xP(x)$ ” or “for every $xP(x)$.” An element for which $P(x)$ is false is called a counterexample to $\forall xP(x)$.

F

2. A compound proposition is satisfiable when it is a contingency.

T

3. A compound proposition is unsatisfiable if and only if its negation is true for all assignments of truth values to the variables.

T

4. The associative law for disjunction shows that the expression $p \vee q \vee r$ is well defined, in the sense that it does not matter whether we first take the disjunction of p with q and then the disjunction of $p \vee q$ with r , or if we first take the disjunction of q and r and then take the disjunction of p with $q \vee r$. It follows that $p_1 \vee p_2 \vee \dots \vee p_n$ are well defined whenever $p_1, p_2, \dots, p_n$ are propositions.

T

5. Proofs in mathematics are valid arguments that establish the truth of mathematical statements.

T

6. An argument is valid if and only if it is impossible for all the premises to be true and the conclusion to be false.

T

7. To deduce new statements from statements we already have, we use rules of inference which are templates for constructing valid arguments.

T

8. When p and q are propositional variables, the statement $((p \rightarrow q) \wedge p) \rightarrow q$ is a tautology. This makes us conclude that when both $p \rightarrow q$ and p are true, q must also be true.

T

9. Suppose that the premises are $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, and $s \rightarrow t$. And the conclusion is t . To show that our premises lead to the desired conclusion, we construct an argument as follows:

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	XXX_1 using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	XXX_2 using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	XXX_3 using (6) and (7)

XXX_1 : Modus tollens

XXX_2 : Modus ponens

XXX_3 : Modus ponens

10. We prove that a conditional statement $p \rightarrow q$ is true when we know that p is false.

T

11. We wish to prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$. by a proof by contraposition. We first assume that the statement $(a \leq \sqrt{n}) \vee (b \leq \sqrt{n})$ is false. Using the meaning of disjunction together with De Morgan's law, we see that this implies that $a > \sqrt{n}$ and $b > \sqrt{n}$. This implies that $ab > n$, which contradicts the statement $n = ab$. Because XXX_1 implies that the hypothesis is false, the original statement is true.

XXX_1 : the negation of the conclusion of the conditional statement

12. In a proof of $p \rightarrow q$ by contraposition, we assume that $\neg q$ is true. We then show that $\neg p$ must also be true. To rewrite a proof by contraposition of $p \rightarrow q$ as a proof by contradiction, we suppose that XXX_1 . Then, we use the steps from the proof of $\neg q \rightarrow \neg p$ to show that $\neg p$ is true. This leads to the contradiction of XXX_2 .

XXX_1 : both p and $\neg q$ are true

XXX_2 : $p \wedge \neg p$

13. To prove a theorem that is a biconditional statement, that is, a statement of the form $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true. The XXX_1 of this is based on the tautology $(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$.

XXX₁ : Validity

14. To prove a statement of the form $(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q$, $[(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q] \leftrightarrow [(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)]$ can be used as a rule of inference. This shows that the original statement with a hypothesis made up of a disjunction of the propositions can be proved by proving each of the conditional statements individually. Such XXX₁ is called a proof by cases.

XXX₁ : (an) argument

15. A valid argument form that can be used in the demonstration that arguments are valid is called XXX₁.

XXX₁ : rule of inference

16. Given two propositions p and q, the proposition, XXX₁, is true when exactly one of p and q is true

XXX₁ : $p \text{ XOR } q$, instead of XOR, symbol circle + also okay.

17. Proof by contradiction can be viewed as a direct proof.

F

18. If we construct the truth table for the proposition $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$, the truth value for TTT, TTF, TFT, TFF, FTT, FTF, FFT, and FFF for p, q, and r corresponds to XXX₁, XXX₂, XXX₃, XXX₄, XXX₅, XXX₆, XXX₇, and XXX₈.

XXX₁ : F

XXX₂ : T

XXX₃ : T

XXX₄ : F

XXX₅ : F

XXX₆ : T

XXX₇ : T

XXX₈ : F

p	q	r	$\neg p$	$\neg r$	$p \leftrightarrow q$	$\neg p \leftrightarrow \neg r$	$(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$
T	T	T	F	F	T	T	F
T	T	F	F	T	T	F	T
T	F	T	F	F	F	T	T
T	F	F	F	T	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	T	F	T	T
F	F	T	T	F	T	F	T
F	F	F	T	T	T	T	F